

Assignment 2:
Object-Oriented Programming (OOP)
CPP602: Java Programming.
Total Marks: 5 Marks

Part A: Classes and Objects
Identify Classes and Objects your project. (1 Mark)

Classes
LeaseAgreement
Invoice

Objects
contract204
marchInvoice

Part B: Encapsulation (1 Mark)
Apply the concept of Encapsulation and state one benefit of using it in your project.

We practice encapsulation by securing the class fields using the private visibility modifier and providing controlled entry through public getter and setter operations.

Benefit
Data security and control: It keeps external methods from altering vital properties directly without checking data validity.

Part C: Inheritance (1 Mark)
Apply Inheritance in your project? Briefly explain how it promotes code reusability.

We can establish a baseline master class called LeaseAgreement that holds common details like agreementId, duration, and monthlyRent.

We then employ inheritance to build derived classes such as CommercialLease and ResidentialLease to manage unique rules.

Instead of repeating identical attributes and methods inside both commercial and residential child structures, they are declared once in the superclass.

The derived subclasses inherit these fields automatically, which clears out duplicate code lines and speeds up deployment.

Part D: Polymorphism (1 Mark)
Apply polymorphism and describe how it allows a program to perform different actions using the same method name.

By utilizing method overriding, the JVM determines during execution exactly which custom variation of a function to trigger depending on the object invoking it.

This enables developers to build uniform transaction flows while allowing separate agreement categories to follow their own distinct processing logic.

Part E: Practical Application

Create a class named Book with the following:

- An attribute called title
- A method called displayTitle()

Write the code to create an object of the class and display the book title.

```
Class Book{
    String title;
    Public void displayTitle() {
        System.out.println("The title of the book is:" + title);
    }
}

Public class main{
    Public static void main (String[] args){
        Book mybook= new Book();
        Mybook.title="Java programming";
        Mybook.displayTitle();
    }
}
```

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